EXACT VILLAINY COMPUTATION AND FIXED-PARAMETER THRESHOLD RECOGNITION

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Editorial note. Complete original mathematical body and embedded bibliography retained; no proof repair, added hypothesis or independent certification.

Editorial note. Source0 omits only pdfTeX-specific glyphtounicode input and pdfgentounicode switch under XeLaTeX. Raw source retained.

Editorial note. Author AI disclosure is retained verbatim in native context: AI assisted developing/checking arguments, experiments, literature and preparation; authors take responsibility. The original does not explicitly declare a core proof discovered by AI, and no further provenance inference is made.

Editorial note. Recognition theorem retains computable-function rather than practical-runtime claims, with exponent independent of threshold and ordinary/weak distinctions intact.

Allowed standard prerequisites (editorial).

- Graph recoloring repair profiles, max-min response composition and Presburger arithmetic. The complete original profile/word definitions, ordinary versus weak repair constraints, transfer and response composition formulas, repeated-union proofs and structural recognition reductions retained.. All native supporting proof components, computational qualification and embedded bibliography included.

1. Introduction

Chromatic villainy measures the cost of repairing a rearrangement of an optimal coloring. The cost counts vertices whose colors change. In the *villainy* $B(G)$ of Clark et al. [2], the repair must preserve every color multiplicity; *weak villainy* $B_{\mathrm{w}}(G)$ permits different multiplicities on the same $\chi(G)$-color palette. Both parameters maximize the minimum repair distance over all such rearrangements. Thus $0 \le B_{\mathrm{w}}(G) \le B(G) \le |V(G)|$. A *word* is any assignment of color labels to vertices, without requiring properness. Its *profile* is the vector of its labeled color multiplicities; it is feasible if a proper coloring realizes it (Theorem 2.1). We call a rearranged source coloring a disturbance.

We study the behavior of these parameters under disjoint union and the complexity of recognizing graphs with a fixed value. Both questions were raised in the early literature; see [2, 14] and the account of Jahanbekam and Lin [12]. Clark et al. proved that weak villainy is superadditive under disjoint union when the summands have the same chromatic number. They proved the ordinary analogue under the additional assumption that every optimal coloring of one summand has equal-sized color classes. It was natural to ask whether these restrictions could be removed, and to determine when equality holds. On the algorithmic side, the classifications of graphs with villainy zero [2] and villainy two [12] give polynomial-time recognition in those cases. The complexity at other fixed values remained to be determined.

This paper continues our study in [13], where we determined odd-cycle profile radii, obtained complete multipartite formulas, and determined the extremal values at fixed order and chromatic number. Here we turn to disjoint unions and computation. The profile viewpoint remains central, but composition requires tracking transfers of color counts and, for ordinary repair, simultaneous responses to different target profiles. Our aim is to describe how global multiplicity constraints combine local repair costs, and to determine when those costs can be computed efficiently. For unions, we retain source profiles and *repair responses*, which record the cost of repairing one source at each target profile (Theorem 4.8). For graphs of small weak villainy, a structural bound reduces all relevant colorings to boundedly many vertex types, grouping vertices with the same neighbors in a specified vertex set (Section 5.2).

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1.1. **Disjoint unions and profile composition.** The requirement that the global source profile be feasible couples the components. Although a disturbance preserves the total number of occurrences of each color, its restriction to a component need not have the profile of any proper coloring of that component. Similarly, an ordinary repair may transfer color multiplicities between components while preserving their global totals. When the chromatic numbers differ, one component also gains access to a larger palette. These features distinguish villainy from the unrestricted Hamming covering radius, the greatest distance of any word to a prescribed set of words. That radius is additive on Cartesian products of codes [3, Section II.C].

Our first results answer both unrestricted superadditivity questions negatively. For every real $C > 0$, there exists a connected graph G such that

$$B_{\mathrm{w}}(G \sqcup K_3) < B_{\mathrm{w}}(G) + B_{\mathrm{w}}(K_3) - C.$$

Thus one summand in the weak counterexample can be a fixed triangle. For ordinary villainy, equal chromatic numbers do not suffice: for every real $C > 0$, there exists a connected graph G with $\chi(G) = 3$ such that

$$B(G \sqcup G) < 2B(G) - C.$$

Theorems 3.1 and 3.2 give explicit constructions.

To describe what replaces additivity, we keep track of source and target profiles. For weak repair, the relevant local information is the largest repair distance among words of each prescribed source profile, with the palette fixed in advance. Theorem 4.3 expresses the value on a union through these local radii. The formula separates the effect of changing the palette from that of transferring color counts between components. It gives an exact equality criterion in local profile data, including when the chromatic numbers differ.

Repeated copies admit a particularly simple weak-repair theorem. Let mG denote the disjoint union of m copies of G, put $k = \chi(G)$, and let $F_k(G)$ be the k-fixing number of Garnero et al. [9]: the maximum distance to a proper k-coloring over *all* source words on that palette. If $m = qk + r$ with $0 \le r < k$, Theorem 4.5 proves

$$0 \le mF_k(G) - B_{\mathrm{w}}(mG) \le r\big(F_k(G) - B_{\mathrm{w}}(G)\big).$$

In particular, $B_{\mathrm{w}}(mG) = mF_k(G)$ whenever k divides m, and $B_{\mathrm{w}}(mG)/m$ converges to $F_k(G)$. Moreover, weak villainy is additive over every repetition if and only if $B_{\mathrm{w}}(G) = F_k(G)$; it suffices to test the union of k copies. The proof balances the color counts of a worst unrestricted source by cyclically relabeling its colors on k copies. Theorem 4.6 makes the eventual behavior effective. For $n = |V(G)|$ and $k \ge 2$, put

$$M = \big(2(k-1)n + 1\big)^{k-1}, \qquad N = (k-1)M.$$

For every $m \ge N$, the defect $mF_k(G) - B_{\mathrm{w}}(mG)$ is periodic. Its least period is

$$h = \gcd\{\ell \ge 1 : B_{\mathrm{w}}(\ell G) = \ell F_k(G)\}, \qquad h \mid k,$$

and the gcd can be computed using only $\ell \le M$. Beyond N, the fixing-number bound is attained exactly when $h \mid m$. These weak-repair results hold on any fixed palette of size at least $\chi(G)$.

Ordinary repair requires more information. A single disturbed coloring has a repair cost for each possible target profile, and these costs must be retained together. Maximizing each coordinate separately can combine costs that no single disturbance realizes. We therefore use the full response functions and discard only those bounded above at every target profile by another realizable response. To combine two responses, we minimize the sum of their costs over all splits of the target profile. This operation, called min-plus convolution in Theorem 4.8, gives Theorem 4.9. The resulting equality tests are finite criteria in component data; they do not constitute a structural classification of all equality pairs.

The response description also yields a general repetition theorem for ordinary villainy. Theorem 4.11 gives a computable rational number $\Lambda(G) \ge F_k(G)$ such that

$$B(mG) = m\Lambda(G) + O_G(1).$$

Here $O_G(1)$ denotes an error bounded by a constant depending only on G. The slope comes from an optimization that allows fractional proportions of copies to receive each source and target coloring, as defined in (4.15). Its maximum is attained by a distribution using at most k source words. Moreover, $B(mG) - m\Lambda(G)$ is eventually periodic, with a computable period and threshold. Thus Theorem 4.12 determines the possible growth of the difference between the repair rules: it is bounded precisely when $\Lambda(G) = F_k(G)$, in which case it is eventually periodic; otherwise it grows linearly.

1.2. **Fixed-parameter recognition.** Our algorithmic result computes both exact parameters using the potentially smaller weak villainy as parameter. Theorem 5.1 gives running time $h(B_{\mathrm{w}}(G))(n+1)^c$, where h is computable and c is an absolute constant, without either value being supplied. Given an integer $t \geq 0$, it also computes the two truncated values

$$\min\{B(G), t+1\} \qquad \text{and} \qquad \min\{B_{\mathrm{w}}(G), t+1\}$$

in time $f(t)(n+1)^c$ for a computable function f. Thus both equality recognition, deciding whether a parameter equals t, and threshold recognition, deciding whether it is at most t, are fixed-parameter tractable in t: the polynomial exponent in the input size is independent of t. They are polynomial-time problems at every fixed value. In particular, unless P = NP, there is no fixed value at which villainy recognition is NP-complete.

The structural input is a matching bound. If $B_{\mathrm{w}}(G) \leq t$, then either G has a vertex cover of size at most $2t$, or $\chi(G) \geq n - 2t$ and the complement of G has a vertex cover of size at most $4t$. The same conclusion holds when $B(G) \leq t$. The proof groups vertices outside these covers by their neighborhoods and retains only source profiles realized by optimal proper colorings. In the small-cover case, a bounded-length formula with variables for the type sizes permits elimination of its quantified variables before the input multiplicities are substituted. In the complement-cover case, symmetries that fix the source coloring remove the need to enumerate singleton color names individually. The polynomial exponent is independent of t; the proof does not give a practical bound on the function f.

1.3. **Related repair problems and organization.** Garnero et al. [9] study repair of an arbitrary given coloring and the fixing number used above. De Biasi and Lauri [5] consider a source promised to be a rearrangement of an optimal coloring. Their r-Fix-Promise problem is an individual weak-repair instance; their swap problem uses the number of color exchanges as cost. Our recognition problem quantifies over every admissible disturbance.

Count-aware component optimization also appears in the two-color repair algorithm of Fellows et al. [7]. It chooses orientations of bipartite components for one given source, subject to total color counts and a swap budget. For villainy, one must additionally retain jointly realizable responses before taking the worst-source maximum.

Finally, endpoint Hamming distance differs from proper-coloring reconfiguration [1, 8], where every intermediate coloring must be proper. Ordinary villainy constrains only the initial and final color multiplicities.

Section 2 fixes the notation and definitions. Section 3 gives the counterexamples to superadditivity. Section 4 develops the profile composition laws and the repeated-copy theorem. Section 5 proves fixed-parameter recognition, and Section 6 discusses further questions.

2. Color profiles and repair distances

All graphs are finite and simple. Unless otherwise stated, a graph G is nonempty, has order n, and has chromatic number k. We use a fixed set $\mathcal{C}$ of k labeled colors.

Definition 2.1 (Words and profiles)**.** A *word* on G is a map $w : V(G) \to \mathcal{C}$, with no properness assumption. Its *profile* is the vector $\mathbf{m}(w) = (m_X(w))_{X \in \mathcal{C}}$, where $m_X(w) = |w^{-1}(X)|$. Bold letters denote vectors. Inequalities between vectors are coordinatewise, $\mathbf{e}_X$ is a coordinate unit vector, and $\mathbf{1}$ is an all-one vector of the required dimension.

Write $\mathcal{P}(G)$ for the proper colorings with palette $\mathcal{C}$. Since $|\mathcal{C}| = \chi(G)$, each such coloring uses every color. A profile $\mathbf{m}$ is *feasible* if it occurs in $\mathcal{P}(G)$.

For a feasible profile, put

$$\mathcal{W}_{\mathbf{m}}(G) = \{w : V(G) \to \mathcal{C} : \mathbf{m}(w) = \mathbf{m}\}, \qquad \mathcal{P}_{\mathbf{m}}(G) = \mathcal{W}_{\mathbf{m}}(G) \cap \mathcal{P}(G).$$

We omit G from $\mathcal{W}_{\mathbf{m}}(G)$ when the vertex set is clear. Every word in $\mathcal{W}_{\mathbf{m}}(G)$ is a rearrangement of every proper coloring with that profile. We call such a word an *admissible disturbance*; equivalently, its profile is feasible. The Hamming distance and the number of agreements are

$$d_{\mathrm{H}}(w, y) = |\{v : w(v) \neq y(v)\}|, \qquad \mathrm{agr}(w, y) = n - d_{\mathrm{H}}(w, y).$$

The cost therefore counts changed vertices; exchanging two different colors on two vertices has cost two.

Definition 2.2 (Repair radii and villainy)**.** The repair radius of a profile and its weak counterpart are

$$R_G(\mathbf{m}) = \max_{w \in \mathcal{W}_{\mathbf{m}}(G)} \min_{y \in \mathcal{P}_{\mathbf{m}}(G)} d_{\mathrm{H}}(w, y),$$
$$R_G^{\mathrm{w}}(\mathbf{m}) = \max_{w \in \mathcal{W}_{\mathbf{m}}(G)} \min_{y \in \mathcal{P}(G)} d_{\mathrm{H}}(w, y).$$

With the terminology of Clark et al. [2],

$$B(G) = \max_{\mathbf{m} \text{ feasible}} R_G(\mathbf{m}), \qquad B_{\mathrm{w}}(G) = \max_{\mathbf{m} \text{ feasible}} R_G^{\mathrm{w}}(\mathbf{m}).$$

Both parameters are zero on the empty graph by convention.

The inclusion of the two target sets gives $B_{\mathrm{w}}(G) \le B(G)$. In particular, the weak target may change the profile, but it still uses the same palette of $\chi(G)$ colors. A simultaneous permutation of the color labels in the source and target preserves distances and all profile conditions.

We write $[k] = \{1, \dots, k\}$. For sets of vectors A, B, their sum is $A + B = \{\mathbf{a} + \mathbf{b} : \mathbf{a} \in A, \mathbf{b} \in B\}$. The graph mG is the disjoint union of m vertex-disjoint copies of G.

3. Counterexamples to superadditivity

Clark et al. [2, Propositions 3.7 and 3.8] proved

$$B_{\mathrm{w}}(G \sqcup H) \ge B_{\mathrm{w}}(G) + B_{\mathrm{w}}(H) \qquad \text{when } \chi(G) = \chi(H), \tag{3.1}$$

and its ordinary analogue when, in addition, every optimal coloring of one summand has equal-sized color classes. We show that the unrestricted inequalities fail by an arbitrarily large amount, even for connected summands.

Theorem 3.1. *For every real $C > 0$, there exists a connected graph G such that*

$$B_{\mathrm{w}}(G \sqcup K_3) < B_{\mathrm{w}}(G) + B_{\mathrm{w}}(K_3) - C.$$

Proof. For a positive integer s, let T_s consist of an edge uv with $3s$ leaves adjoined to each endpoint. Both bipartition classes have size $3s + 1$. The connected bipartite formula [2, Proposition 3.3] gives

$$B_{\mathrm{w}}(T_s) = 2 \left\lfloor \frac{3s+1}{2} \right\rfloor, \qquad B_{\mathrm{w}}(K_3) = 0.$$

We show that $B_{\mathrm{w}}(T_s \sqcup K_3) \le 2s + 3$.

The union has chromatic number three. For any three-color source w, let $N_u(i), N_v(i)$ count the leaves of color i at the two centers. Repairing the center colors to distinct labels i, j has minimum tree cost

$$N_u(i) + N_v(j) + \mathbf{1}_{\{w(u) \ne i\}} + \mathbf{1}_{\{w(v) \ne j\}}.$$

Its average over the six ordered pairs is $2s + 4/3$. Some integer cost is therefore at most $2s + 1$. The triangle needs at most two changes, giving the required upper bound.

It follows that

$$B_{\mathrm{w}}(T_s) + B_{\mathrm{w}}(K_3) - B_{\mathrm{w}}(T_s \sqcup K_3) \ge 2 \left\lfloor \frac{3s+1}{2} \right\rfloor - (2s + 3) \ge s - 3.$$

Choosing an integer $s > C + 3$ proves the stated strict inequality. □

For the ordinary parameter, the chromatic numbers can be equal and the two connected summands can even be isomorphic. We use the following general observation about a feasible profile $\mathbf{p}$ on N vertices. If two proper targets of that profile have disjoint color-X classes of size p_X, then every source of profile $\mathbf{p}$ agrees with one of them on at least

$$\max \left\{ 0, \left\lceil \frac{3p_X - N}{2} \right\rceil \right\} \tag{3.2}$$

vertices. Indeed, the source X-set has size p_X, and its intersection with the union of the two target X-sets has size at least $3p_X - N$. One of the two intersections has at least half that size.

Theorem 3.2. *For every real $C > 0$, there exists a connected graph G with $\chi(G) = 3$ such that*

$$B(G \sqcup G) < 2B(G) - C.$$

Here the union consists of two vertex-disjoint copies of G.

Proof. For each positive integer t, take

$$J_t = \overline{K}_{5t} \vee \big(K_{t,2t} \sqcup K_{t,3t}\big),$$

where the join adds every edge between its two arguments. This is a connected graph of order $12t$ and chromatic number three. Denote its five independent vertex sets by S, L_1, R_1, L_2, R_2, of respective sizes $5t, t, 2t, t, 3t$. The set S is joined to all four other sets, and the remaining edges are those between L_i and R_i for $i = 1, 2$.

In a proper three-coloring, the bipartite remainder requires two colors, so S must use only the third. Each of its two connected components then has its two possible binary colorings. The possible unordered profiles on J_t are precisely $t(2, 5, 5)$ and $t(3, 4, 5)$. The labeled profile $(3t, 4t, 5t)$ has a unique proper realization: C occupies S, A occupies $R_1 \cup L_2$, and B occupies $L_1 \cup R_2$. A source with this profile and no agreement is described by the matrix

$$t \begin{pmatrix} 0 & 2 & 1 \\ 0 & 0 & 4 \\ 3 & 2 & 0 \end{pmatrix},$$

whose rows are these target classes and whose columns count the source colors. Its row and column sums are $(3t, 4t, 5t)$, and its diagonal vanishes. Hence $B(J_t) = 12t$.

Adding two permutations of the possible profiles gives exactly the following sorted profiles on $2J_t$, divided by t:

$$(4, 10, 10),\ (5, 9, 10),\ (6, 8, 10),\ (6, 9, 9),\ (7, 7, 10),\ (7, 8, 9),\ (8, 8, 8).$$

For clarity, adding two copies of $(2, 5, 5)$ gives the first or fifth; mixing the two types gives the second, third, or sixth; adding two copies of $(3, 4, 5)$ gives the third through seventh. In every unbalanced case the table supplies two targets with disjoint C-classes. Each five-letter word assigns colors to (S, L_1, R_1, L_2, R_2) in one copy, and the separator distinguishes the two copies.

Profile divided by t	First target	Second target
$(4, 10, 10)$	$BACAC \mid BACAC$	$CABAB \mid CABAB$
$(5, 9, 10)$	$BACAC \mid CBAAB$	$CBAAB \mid BACAC$
$(6, 8, 10)$	$BACAC \mid CABBA$	$CABBA \mid BACAC$
$(6, 9, 9)$	$BCAAC \mid CBAAB$	$CBAAB \mid BCAAC$
$(7, 7, 10)$	$ABCBC \mid BACAC$	$CABAB \mid CBABA$
$(7, 8, 9)$	$ACBBC \mid CABAB$	$CABAB \mid ACBBC$

Every word is proper: its first letter differs from the other four, its second differs from its third, and its fourth differs from its fifth. Weighted letter counts verify each profile. In no position do both targets have C, so their C-classes are disjoint.

Here $N = 24t$ and $p_C \geq 9t$. By (3.2), every source of an unbalanced profile has a repair retaining at least $\lceil 3t/2 \rceil$ vertices. For the balanced profile $(8t, 8t, 8t)$, the three cyclic relabelings of any proper target still have that profile and together agree once at every vertex. Some target therefore retains at least $8t$ vertices. Consequently

$$B(2J_t) \leq 24t - \left\lceil \frac{3t}{2} \right\rceil, \qquad 2B(J_t) - B(2J_t) \geq \left\lceil \frac{3t}{2} \right\rceil.$$

Choose an integer $t > 2C/3$. Then $\lceil 3t/2 \rceil > C$, which gives the stated strict inequality. The upper bound on $B(2J_t)$ is all that is needed here; no exact value for it is asserted. □

4. Profile composition and repeated unions

The counterexamples in Section 3 show the effect of permitting color counts to move between components. We now retain the local profile data and derive exact composition laws. For repeated copies, these laws give eventual linear growth with periodic corrections under both repair rules. Weak repair has the explicit slope $F_k(G)$; ordinary repair has a rational slope described by a finite fractional optimization.

4.1. **Weak repair and transfer of color counts.** To analyze arbitrary summands, their colorings must first be expressed in a common palette.

Definition 4.1 (Common-palette radii)**.** Fix $k \geq \chi(J)$, and let $\mathcal{P}_k(J)$ be the set of proper maps from $V(J)$ to $[k]$. These maps may leave some labels unused. Write

$$\mathcal{A}_k(J) = \left\{ \mathbf{a} \in \mathbb{Z}^k_{\geq 0} : \sum_{i=1}^{k} a_i = |V(J)| \right\}, \qquad \mathcal{M}_k(J) = \{\mathbf{m}(y) : y \in \mathcal{P}_k(J)\}.$$

Thus $\mathcal{A}_k(J)$ contains all source profiles, whereas $\mathcal{M}_k(J)$ contains the profiles realized by proper colorings. The distinction matters even when $k = \chi(J)$: a source restricted to one summand of a union need not have a feasible profile on that summand.

For every $\mathbf{a} \in \mathcal{A}_k(J)$, define

$$\rho_{J,k}(\mathbf{a}) = \max_{\substack{w:V(J)\to[k] \\ \mathbf{m}(w)=\mathbf{a}}} \min_{y \in \mathcal{P}_k(J)} d_{\mathrm{H}}(w, y).$$

We shall use the two maxima

$$\beta_k(J) = \max_{\mathbf{p} \in \mathcal{M}_k(J)} \rho_{J,k}(\mathbf{p}), \qquad F_k(J) = \max_{\mathbf{a} \in \mathcal{A}_k(J)} \rho_{J,k}(\mathbf{a}). \tag{4.1}$$

The first is weak villainy with the fixed palette $[k]$; in particular, $\beta_{\chi(J)}(J) = B_{\mathrm{w}}(J)$. The second is the k-fixing number studied by Garnero et al. [9], in which the source coloring is unrestricted. Put

$$\varepsilon_k(J) = F_k(J) - \beta_k(J) \geq 0.$$

This difference measures how much is lost by requiring the source profile to be feasible.

Definition 4.2 (Transfer surplus)**.** For an integral vector $\boldsymbol{\delta}$ whose coordinates sum to zero, define the *transfer surplus*

$$\Delta_{J,k}(\boldsymbol{\delta}) = \max_{\substack{\mathbf{p} \in \mathcal{M}_k(J) \\ \mathbf{p}+\boldsymbol{\delta} \geq \mathbf{0}}} \rho_{J,k}(\mathbf{p} + \boldsymbol{\delta}) - \beta_k(J). \tag{4.2}$$

An empty maximum has value $-\infty$. The vector $\boldsymbol{\delta}$ records a change in the local color counts relative to a proper coloring. Its coordinates can have both signs, although their sum is zero because the order of the summand is unchanged. For a union, this change must be offset by $-\boldsymbol{\delta}$ on the other summand.

Theorem 4.3. *Let $k \geq \max\{\chi(G), \chi(H)\}$, and put*

$$T_k(G, H) = \max_{\boldsymbol{\delta}} \big(\Delta_{G,k}(\boldsymbol{\delta}) + \Delta_{H,k}(-\boldsymbol{\delta}) \big).$$

Then

$$\beta_k(G \sqcup H) = \beta_k(G) + \beta_k(H) + T_k(G, H), \tag{4.3}$$

where

$$0 \leq T_k(G, H) \leq \varepsilon_k(G) + \varepsilon_k(H). \tag{4.4}$$

Proof. A proper coloring of $G \sqcup H$ is exactly a pair of local proper colorings, and hence

$$\mathcal{M}_k(G \sqcup H) = \mathcal{M}_k(G) + \mathcal{M}_k(H).$$

For fixed local source words, the two weak repairs are independent and their costs add. Maximizing over sources of prescribed local profiles therefore gives

$$\beta_k(G \sqcup H) = \max_{\substack{\mathbf{a} \in \mathcal{A}_k(G), \ \mathbf{c} \in \mathcal{A}_k(H) \\ \mathbf{a}+\mathbf{c} \in \mathcal{M}_k(G)+\mathcal{M}_k(H)}} \big(\rho_{G,k}(\mathbf{a}) + \rho_{H,k}(\mathbf{c}) \big). \tag{4.5}$$

For every admitted pair $\mathbf{a}, \mathbf{c}$, choose proper profiles $\mathbf{p}, \mathbf{q}$ with $\mathbf{p} + \mathbf{q} = \mathbf{a} + \mathbf{c}$. Then

$$\boldsymbol{\delta} = \mathbf{a} - \mathbf{p} = \mathbf{q} - \mathbf{c}$$

has coordinate sum zero. Conversely, proper profiles $\mathbf{p}, \mathbf{q}$ and a vector $\boldsymbol{\delta}$ for which $\mathbf{p} + \boldsymbol{\delta}$ and $\mathbf{q} - \boldsymbol{\delta}$ are nonnegative determine an admitted source pair. For fixed $\boldsymbol{\delta}$, the choices of $\mathbf{p}$ and $\mathbf{q}$ are independent. Substitution into (4.5) and separate maximization over these two profiles prove (4.3).

The zero transfer satisfies $\Delta_{J,k}(\mathbf{0}) = 0$, so $T_k(G, H) \geq 0$. Also $\rho_{J,k}(\mathbf{p} + \boldsymbol{\delta}) \leq F_k(J)$, which bounds each surplus by $\varepsilon_k(J)$ and proves the upper bound in (4.4). □

The theorem identifies two distinct effects in the original superadditivity question. A common palette changes the individual weak-villainy values to $\beta_k(G)$ and $\beta_k(H)$. Relative to these common-palette values, transferring color counts can only increase the worst repair cost, and its entire contribution is $T_k(G,H)$.

Corollary 4.4. *If $\chi(G)=\chi(H)=k$, then*

$$B_{\mathrm{w}}(G\sqcup H)=B_{\mathrm{w}}(G)+B_{\mathrm{w}}(H)$$

if and only if

$$\Delta_{G,k}(\boldsymbol{\delta})+\Delta_{H,k}(-\boldsymbol{\delta})\le 0 \qquad \text{for every } \boldsymbol{\delta}. \tag{4.6}$$

In particular, equality holds if $F_k(G)=B_{\mathrm{w}}(G)$ and $F_k(H)=B_{\mathrm{w}}(H)$.

For arbitrary G,H, set $k=\max\{\chi(G),\chi(H)\}$ and

$$\Gamma_k(G,H)=B_{\mathrm{w}}(G)+B_{\mathrm{w}}(H)-\beta_k(G)-\beta_k(H).$$

Then

$$B_{\mathrm{w}}(G\sqcup H)-B_{\mathrm{w}}(G)-B_{\mathrm{w}}(H)=T_k(G,H)-\Gamma_k(G,H). \tag{4.7}$$

Consequently superadditivity holds if and only if $T_k(G,H)\ge\Gamma_k(G,H)$, and equality holds if and only if $T_k(G,H)=\Gamma_k(G,H)$.

Proof. At equal chromatic numbers the palette correction is zero, and the maximum defining T_k already contains the zero transfer. Thus $T_k=0$ is equivalent to (4.6). The sufficient condition follows from (4.4). For unequal chromatic numbers, subtract the two intrinsic weak-villainy values from (4.3). □

These criteria use local profile radii and remain valid for every graph pair. The correction $\Gamma_k(G,H)$ is kept signed; the argument does not require an assumption about how the feasible-source maximum changes when additional colors are made available.

4.2. **Repeated copies and the fixing number.** The unrestricted maximum $F_k(G)$ may be attained by a word whose profile is not feasible. Taking several copies of the graph permits such a word to be balanced across the palette. This gives an exact relation between weak villainy and the fixing number.

Theorem 4.5. *Fix an integer $k\ge\max\{1,\chi(G)\}$, and write $m=qk+r$, where $q\ge 0$ and $0\le r<k$. Then*

$$0\le mF_k(G)-\beta_k(mG)\le r\big(F_k(G)-\beta_k(G)\big). \tag{4.8}$$

In particular,

$$\beta_k(mG)=mF_k(G) \qquad \text{whenever } k\mid m, \tag{4.9}$$

and

$$\lim_{m\to\infty}\frac{\beta_k(mG)}{m}=F_k(G). \tag{4.10}$$

Proof. Every source on mG can be weakly repaired one copy at a time, each at cost at most $F_k(G)$. This proves the upper bound $\beta_k(mG)\le mF_k(G)$.

Let w be a word on G at distance $F_k(G)$ from $\mathcal{P}_k(G)$, and let π be a cyclic permutation of all k color labels. On k copies of G, use the sources

$$w,\ \pi\circ w,\ \dots,\ \pi^{k-1}\circ w.$$

If G has n vertices, their total profile is $(n,\dots,n)$: the k copies of every vertex receive every label once. This profile is feasible. Indeed, starting with any proper $[k]$-coloring y of G, its k cyclic relabelings give a proper coloring of kG with the same total profile. Relabeling preserves distances to the proper-coloring set, so each of the k sources still requires $F_k(G)$ changes in a weak repair. The repairs on distinct copies are independent. Thus this admitted source on kG requires exactly $kF_k(G)$ changes.

Repeating the construction gives qk copies with cost $qkF_k(G)$. On each of the remaining r copies, use a feasible source attaining $\beta_k(G)$. The resulting total profile is feasible, since every block and each remaining copy has a feasible profile. Its weak repair cost is

$$qkF_k(G)+r\beta_k(G).$$

Combining this lower bound with the upper bound proves (4.8). The exact formula when $r=0$ follows immediately, and the error in (4.8) is bounded independently of m, proving (4.10). □

When $k = \chi(G)$, the common-palette quantities in this theorem are the intrinsic weak-villainy values.

The error in equation (4.8) eventually repeats. The following result gives an explicit stabilization bound and identifies its least period from the repetitions attaining the fixing-number bound.

Theorem 4.6 (Effective weak repetition)**.** *Let G have n vertices and let $k \geq \max\{2, \chi(G)\}$. Put*

$$F = F_k(G), \qquad \varepsilon = F - \beta_k(G), \qquad M = \big(2(k-1)n + 1\big)^{k-1}, \qquad N = (k-1)M.$$

Define

$$h = \gcd\{\ell \geq 1 : \beta_k(\ell G) = \ell F\}.$$

Then $h \mid k$, and there are integers $c_0, \ldots, c_{h-1}$ such that

$$c_0 = 0, \qquad 1 \leq c_r \leq r\varepsilon \quad (1 \leq r < h), \qquad \beta_k(mG) = mF - c_{m \bmod h} \quad (m \geq N).$$

The number h is the least eventual period of the defect sequence $d_m = mF - \beta_k(mG)$: it is the least positive integer for which $d_{m+h} = d_m$ for all sufficiently large m. In particular, for $m \geq N$,

$$\beta_k(mG) = mF \quad \Longleftrightarrow \quad h \mid m.$$

Moreover,

$$h = \gcd\{1 \leq \ell \leq M : \beta_k(\ell G) = \ell F\}.$$

The period and coefficients are computable from the feasible profiles and the common-palette radii of G.

Proof. Write $d = k - 1$. We use the Steinitz lemma in the following standard form: vectors in $\mathbb{R}^d$ of ℓ_∞ norm at most n and with sum zero can be ordered so that every partial sum has ℓ_∞ norm at most dn; see [6]. Consequently a nonempty zero-sum multiset of integer vectors of norm at most n that has no proper nonempty zero-sum submultiset contains at most $(2dn+1)^d = M$ vectors. Indeed, for a multiset of length L, the partial sums $\mathbf{s}_0 = \mathbf{0}, \mathbf{s}_1, \ldots, \mathbf{s}_{L-1}$ in the Steinitz ordering are distinct lattice points in $[-dn, dn]^d$. An equality between two of these sums would produce a proper nonempty zero-sum submultiset.

To each pair $(\mathbf{a}, \mathbf{p}) \in \mathcal{A}_k(G) \times \mathcal{M}_k(G)$, associate the vector

$$\boldsymbol{\delta} = (a_1 - p_1, \ldots, a_d - p_d)$$

and nonnegative integer cost $F - \rho_{G,k}(\mathbf{a})$. The omitted coordinate of $\mathbf{a} - \mathbf{p}$ is determined by the first d, because its coordinate sum is zero. Thus a multiset of m local pairs has vector sum zero precisely when its source profiles sum to its proper profiles. Such a choice is exactly a feasible global source profile on mG. For fixed local source profiles, the maximum weak repair cost is the sum of the local radii, since sources can be chosen independently and weak repairs on different copies are independent. It follows that

$$d_m := mF - \beta_k(mG)$$

is the minimum total cost of a zero-sum multiset of m local pairs. The length of a multiset is its number of pairs, counted with multiplicity, and its cost is the sum of their costs. We also allow the empty multiset and put $d_0 = 0$.

Call a nonempty zero-sum multiset a block if it has no proper nonempty zero-sum submultiset. Every zero-sum multiset is a disjoint union of blocks, by successively taking a minimal nonempty zero-sum submultiset. Each block has length at most M, by the preceding Steinitz consequence. There are only finitely many blocks because the set of local pairs is finite.

Consider the directed graph with vertex set $\mathbb{Z}/k\mathbb{Z}$. For every block of length ℓ and cost b, place an edge from each residue u to $u + \ell$, with weight b. A path starting at zero represents a concatenation of blocks, so the shortest-path distance a_r from zero to r is

$$a_r = \min_{m \geq 0,\ m \equiv r \pmod{k}} d_m.$$

Every residue is reachable: a local pair with $\mathbf{a} = \mathbf{p}$ is itself a block of length one. Edge weights are nonnegative, so a shortest path can be chosen without repeated vertices. It has at most $k - 1$ edges and therefore represents a witness of length $m_r \leq (k-1)M = N$ and cost a_r.

The cyclic relabeling construction in Theorem 4.5 gives a zero-cost multiset of length k. For any $m \geq N$ with $m \equiv r \pmod{k}$, append $(m - m_r)/k$ copies of that multiset to the witness above. This gives $d_m \leq a_r$, while the definition of a_r gives the reverse inequality. Hence

$$d_m = a_{m \bmod k} \qquad (m \geq N). \tag{4.11}$$

The original error bound also gives $a_0 = 0$ and $a_r \le r\varepsilon$ for $0 \le r < k$.

To identify the least period, let H be the subgroup of $\mathbb{Z}/k\mathbb{Z}$ generated by the lengths of zero-cost blocks. Nonnegative edge weights imply that $a_r = 0$ exactly when r is reachable from zero using only zero-cost edges. In a finite group, the semigroup generated by given residues is their subgroup; thus $a_r = 0$ exactly when $r \in H$. Appending a zero-cost path shows that $a_{r+u} \le a_r$ for $u \in H$. Applying the same argument with $-u \in H$ proves equality. Consequently (a_r) is constant on the cosets of H.

The zero-cost multiset of length k decomposes into zero-cost blocks, because all costs are nonnegative. Hence the gcd of the lengths of zero-cost blocks divides k. Every zero-cost multiset also decomposes into zero-cost blocks. Its possible lengths are therefore precisely the additive semigroup generated by those block lengths, that is, their nonnegative integer combinations, and their gcd is the number h in the statement. Thus $H = h\mathbb{Z}/k\mathbb{Z}$. The sequence (a_r) has period h. Any period t must send its zero entry a_0 to another zero entry, so $t \bmod k \in H$; hence h is its least positive period. By (4.11), it is also the least eventual period of (d_m).

Set $c_r = a_r$ for $0 \le r < h$. These coefficients satisfy the stated bounds, and $c_r > 0$ when $r \neq 0$ because such a residue does not belong to H. Every zero-cost block has length at most M and is itself a zero-cost multiset. Conversely, every zero-cost multiset of length at most M decomposes into zero-cost blocks. The two collections of lengths have the same gcd, proving the finite formula for h.

Finally, all local pairs, their costs, and all blocks of length at most M can be enumerated from the displayed local data. The shortest-path construction then computes the coefficients, and the zero-cost block lengths compute h. This is a finite procedure; no polynomial bound on the enumeration is asserted. □

When $k = 1$, the graph is edgeless and $F_1(G) = \beta_1(mG) = 0$.

Corollary 4.7. *For a graph G with $\chi(G) = k$, the following are equivalent:*

(1) $B_{\mathrm{w}}(G) = F_k(G)$*;*

(2) $B_{\mathrm{w}}(mG) = mB_{\mathrm{w}}(G)$ *for every positive integer m;*

(3) $B_{\mathrm{w}}(kG) = kB_{\mathrm{w}}(G)$.

Proof. If the first equality holds, the error bound in Theorem 4.5 is zero for every m. The second statement includes the third. Finally, (4.9) with $m = k$ shows that the third implies the first. □

Thus additivity over every repetition is equivalent to a property of a single graph: an unrestricted worst source can be chosen with a feasible profile.

4.3. **Count-preserving repairs and response functions.** For ordinary villainy, the target profiles on the summands are coupled by their prescribed sum. We retain the common palette of k colors and the profile sets $\mathcal{A}_k(J), \mathcal{M}_k(J)$ introduced in Theorem 4.1. Local source profiles range over $\mathcal{A}_k(J)$, even when they are not proper profiles.

Definition 4.8 (Repair responses)**.** The *response function* of a word $w : V(J) \to [k]$ of profile $\mathbf{a}$ is

$$f_w(\mathbf{p}) = \min_{\substack{y \in \mathcal{P}_k(J) \\ \mathbf{m}(y) = \mathbf{p}}} d_{\mathrm{H}}(w, y) \qquad (\mathbf{p} \in \mathcal{M}_k(J)).$$

Here all colorings and words use the common palette; a proper coloring of a summand need not use every color. Extend f_w by $+\infty$ outside $\mathcal{M}_k(J)$. Write

$$\mathcal{R}_k(J, \mathbf{a}) = \mathrm{Max}\{f_w : \mathbf{m}(w) = \mathbf{a}\},$$

where Max retains the coordinatewise maximal functions and removes duplicates. Thus a response is discarded if another realizable response is at least as large at every target profile and larger at some profile. The resulting family is an *antichain*: no two distinct remaining responses are comparable coordinatewise. It records the possible tradeoffs between target profiles without retaining the source words themselves.

For two response functions define their min-plus convolution by

$$(f \star g)(\mathbf{p}) = \min_{\mathbf{u}+\mathbf{v}=\mathbf{p}} \big(f(\mathbf{u}) + g(\mathbf{v})\big).$$

The profiles where its value is finite are exactly $\mathcal{M}_k(G) + \mathcal{M}_k(H)$ when f, g are responses on G, H, respectively. For sets of responses, put $\mathcal{U} \star \mathcal{V} = \{f \star g : f \in \mathcal{U}, g \in \mathcal{V}\}$.

Theorem 4.9. *For every $k \geq \max\{\chi(G), \chi(H)\}$ and every source profile $\mathbf{a}$ on $G \sqcup H$,*

$$\mathcal{R}_k(G \sqcup H, \mathbf{a}) = \operatorname{Max} \bigcup_{\boldsymbol{\alpha}+\boldsymbol{\beta}=\mathbf{a}} \big(\mathcal{R}_k(G, \boldsymbol{\alpha}) \star \mathcal{R}_k(H, \boldsymbol{\beta})\big), \tag{4.12}$$

where the union ranges over $\boldsymbol{\alpha} \in \mathcal{A}_k(G)$ and $\boldsymbol{\beta} \in \mathcal{A}_k(H)$. This composition is associative and commutative.

Proof. A source word on the union restricts to words w, z on the two summands. At a fixed target profile $\mathbf{p}$, any proper repair has local target profiles $\mathbf{u}, \mathbf{v}$ with $\mathbf{u} + \mathbf{v} = \mathbf{p}$. Once these profiles are fixed, the two repairs are independent and their distances add. The union response is therefore exactly $f_w \star f_z$. Conversely, any pair of local response functions can be realized simultaneously by choosing its two source words independently.

This proves (4.12) before dominated responses are removed. Convolution is monotone in each argument: if $f \leq f'$ and $g \leq g'$, then $f \star g \leq f' \star g'$. Replacing a discarded local response by a maximal one can consequently only increase the resulting response. Pruning the local lists thus leaves the union's maximal responses unchanged. Associativity and commutativity follow from those of min-plus convolution, and the same monotonicity permits pruning at every intermediate composition. □

This theorem is a reduction to finite local data; it does not assert that the response antichains are always small. When $k = \chi(J)$, ordinary villainy is recovered from their diagonal entries:

$$B(J) = \max_{\mathbf{a} \in \mathcal{M}_k(J)} \max_{f \in \mathcal{R}_k(J, \mathbf{a})} f(\mathbf{a}). \tag{4.13}$$

The antichain is a singleton whenever one source attains every coordinatewise maximum simultaneously. The common extremizer in [13, Theorem 3.15] gives

$$\mathcal{R}_3(C_{2r+1}, \mathbf{a}) = \{f_{\mathbf{a}}\},$$
$$f_{\mathbf{a}}(\mathbf{p}) = 2r + 1 - \sum_{X \in [3]} \min\{p_X, \lceil a_X/2 \rceil\} \qquad (\mathbf{p} \in \mathcal{M}_3(C_{2r+1}))$$

for every source profile $\mathbf{a}$, including infeasible ones: the same cyclic word, with all occurrences of each color consecutive, attains the displayed value at every target profile. Thus these directed radii supply the complete local response data needed for composition.

In general, however, it is essential to retain the antichain, rather than merely the largest value in each coordinate. For example, on $P_3 = K_{1,2}$ with two colors, order the target profiles by their A-counts $1, 2$. Sources with one A have responses $(0, 3)$ and $(2, 1)$, according as A occupies the center or a leaf. Sources with two A's have responses $(3, 0)$ and $(1, 2)$. For two copies and the source split $1 + 2$, the greatest convolution value at total A-count three is two. Taking coordinatewise maxima first instead gives

$$\min\{2 + 2, 3 + 3\} = 4.$$

Indeed, the actual radius of profile $(3, 3)$ on $2P_3$ is two: its two proper targets are complementary, their distances from a same-profile source are even and sum to six, and a source constant on each copy attains two. Thus the outer maximum over disturbances cannot in general be interchanged with the minimum over target profiles.

The response functions also give an exact criterion for equality under disjoint union. For an integer j, put

$$\mathcal{L}_f(j) = \{\mathbf{p} : f(\mathbf{p}) \leq j\}, \qquad \mathcal{K}_L(f, g) = \bigcup_{j=0}^{L} \big(\mathcal{L}_f(j) + \mathcal{L}_g(L - j)\big),$$

with $\mathcal{L}_f(j) = \varnothing$ for $j < 0$ and $\mathcal{K}_{-1}(f, g) = \varnothing$.

Proposition 4.10. *Set $k = \chi(G \sqcup H)$ and let $L \geq 0$ be an integer. Then $B(G \sqcup H) \leq L$ if and only if*

$$\boldsymbol{\alpha} + \boldsymbol{\beta} \in \mathcal{K}_L(f, g) \tag{4.14}$$

for every $f \in \mathcal{R}_k(G, \boldsymbol{\alpha})$ and $g \in \mathcal{R}_k(H, \boldsymbol{\beta})$ whose summed source profile belongs to $\mathcal{M}_k(G) + \mathcal{M}_k(H)$. In particular, for $L = B(G) + B(H)$, equality $B(G \sqcup H) = B(G) + B(H)$ holds if and only if this condition holds for every such pair at budget L, and fails for at least one such pair at budget $L - 1$.

Proof. For a fixed pair of responses, $(f \star g)(\mathbf{m}) \leq L$ means that $\mathbf{m} = \mathbf{p} + \mathbf{q}$ for profiles satisfying $f(\mathbf{p}) + g(\mathbf{q}) \leq L$. Since costs are integers, this is equivalent to $\mathbf{m} \in \mathcal{K}_L(f, g)$. Theorem 4.9 and (4.13) now give the asserted upper-bound criterion. Failure at $L - 1$, together with the bound at L, is equivalent to equality. The argument does not require equal chromatic numbers for the summands. □

This is an exact compositional criterion in local profile data, rather than an elementary structural classification of the graph pairs. It distinguishes two effects: target multiplicities may transfer between summands during repair, and a globally feasible disturbance may have locally infeasible profiles.

4.4. **The growth of ordinary villainy under repetition.** The response composition law also determines the asymptotic behavior of ordinary villainy. Fix a graph G with $\chi(G) = k$, and let $\mathcal{W}$ and $\mathcal{P}$ be, respectively, all words and all proper colorings of G on $[k]$. Both sets are finite. A vector $\mathbf{x} = (x_w)_{w\in\mathcal{W}}$ describes proportions of source words on many copies of G. Its average profile must belong to the convex hull of the proper profiles. Accordingly, put

$$\mathcal{X}_G = \left\{ \mathbf{x} \geq \mathbf{0} : \sum_w x_w = 1, \quad \sum_w x_w \mathbf{m}(w) \in \operatorname{conv} \mathcal{M}_k(G) \right\}.$$

For $\mathbf{x} \in \mathcal{X}_G$, define the fractional repair cost

$$\begin{aligned} \Phi_G(\mathbf{x}) = \min_{\mathbf{z}\geq\mathbf{0}} \quad & \sum_{w\in\mathcal{W},\ y\in\mathcal{P}} d_{\mathrm{H}}(w,y) z_{wy}, \\ \text{subject to} \quad & \sum_y z_{wy} = x_w \quad (w \in \mathcal{W}), \\ & \sum_{w,y} z_{wy}\big(\mathbf{m}(y) - \mathbf{m}(w)\big) = \mathbf{0}. \end{aligned} \tag{4.15}$$

The variable z_{wy} assigns a proportion of copies with source w to the proper target y. Thus the second constraint preserves the total color counts. The program is feasible precisely for $\mathbf{x} \in \mathcal{X}_G$. Set

$$\Lambda(G) = \max_{\mathbf{x}\in\mathcal{X}_G} \Phi_G(\mathbf{x}). \tag{4.16}$$

Theorem 4.11 (Ordinary repetition)**.** *For every fixed graph G with $k = \chi(G)$, the number $\Lambda(G)$ is rational and effectively computable. The maximum in (4.16) is attained by a distribution supported on at most k source words, meaning that at most k coordinates x_w are positive. Moreover,*

$$B(mG) = m\Lambda(G) + O_G(1), \qquad \Lambda(G) \geq F_k(G). \tag{4.17}$$

The error term is bounded by a constant depending only on G. There are effectively computable integers $h \geq 1$, $a \geq 0$, and $M \geq 1$ such that $a = h\Lambda(G)$ and

$$B((m+h)G) = B(mG) + a \qquad (m \geq M). \tag{4.18}$$

Proof. Write the constraints in (4.15) as $A\mathbf{z} = (\mathbf{x}, \mathbf{0})$, where A is a fixed integer matrix, and write $\mathbf{c}$ for the nonnegative integer cost vector. When $\mathbf{x} \in \mathbb{Z}_{\geq 0}^{\mathcal{W}}$ counts actual source copies, its sum is their number. The source is feasible if and only if the same system has a nonnegative integer solution. In that case the minimum over integer solutions, denoted $C_G(\mathbf{x})$, is its ordinary repair cost. This is the repeated form of Theorem 4.9. For unnormalized vectors, we define Φ_G by the same program; scaling all source proportions by a nonnegative factor scales its optimum by that factor.

For this fixed matrix and cost vector, the integer-programming proximity bound of Cook, Gerards, Schrijver, and Tardos [4, Corollary 2] gives a constant D, independent of $\mathbf{x}$, such that

$$0 \leq C_G(\mathbf{x}) - \Phi_G(\mathbf{x}) \leq D \tag{4.19}$$

whenever an integer solution exists. To apply that bound, write the equalities as pairs of inequalities and include nonnegativity. The resulting integral constraint matrix and objective are fixed, and the row sums bound every feasible variable.

The function Φ_G is convex and piecewise linear by linear programming duality. Its maximum on the rational polytope $\mathcal{X}_G$ is attained at a vertex. Such a vertex has at most k positive coordinates: more than k corresponding profile vectors would be linearly dependent, and a sufficiently small perturbation in that dependence would preserve both the average profile and the sum of the coordinates. Enumerating the vertices and solving the corresponding rational linear programs computes a rational value $\Lambda(G)$. Any feasible integral source count vector with sum m has $\mathbf{x}/m \in \mathcal{X}_G$. Thus (4.19) gives $B(mG) \leq m\Lambda(G) + D$.

For the reverse bound, choose a rational maximizer $\mathbf{x}^* \in \mathcal{X}_G$. Express its average profile as a rational convex combination of proper profiles, and choose a common denominator d for these coefficients and $\mathbf{x}^*$. Then $d\mathbf{x}^*$ is the source count vector of a feasible source on dG: the resulting integer numbers of source copies and proper target copies have the same total profile and may be paired arbitrarily. Fix a proper word y_0. For $m = qd + r$, $0 \leq r < d$, the integer vector $\mathbf{x}^{(m)} = qd\mathbf{x}^* + r\mathbf{e}_{y_0}$ is feasible. Take an

optimal dual solution at $\mathbf{x}^*$, and let $\mathbf{u}$ denote its coordinates corresponding to the row-sum constraints. Dual feasibility holds for every source count vector, so

$$B(mG) \geq C_G(\mathbf{x}^{(m)}) \geq \Phi_G(\mathbf{x}^{(m)}) \geq qd\Lambda(G) + ru_{y_0} = m\Lambda(G) - r\big(\Lambda(G) - u_{y_0}\big).$$

The final error is bounded independently of m, proving the first assertion of (4.17). The second follows from $B(mG) \geq B_{\mathrm{w}}(mG)$ and Theorem 4.5.

Finally, the set of pairs (m, b) with $b = B(mG)$ is definable in Presburger arithmetic, the language of integer linear inequalities and congruences combined by Boolean operations and quantifiers. Indeed, feasible sources, integer repairs, and their costs are described by the fixed linear system above. Let $P(m, t)$ assert that some feasible integer source count vector of sum m has every integer repair of cost at least t. This is a Presburger formula, and $B(mG) = b$ is expressed by $P(m, b) \wedge \neg P(m, b + 1)$. Write $\Lambda(G) = a_0/h_0$ with integers $a_0 \geq 0$ and $h_0 \geq 1$. By (4.17), the integer $h_0 B(mG) - a_0 m$ belongs to a fixed finite set. For each of its possible values, the set of corresponding m is a one-variable Presburger set, hence effectively eventually periodic by effective quantifier elimination [10, 11], which replaces a quantified formula by an equivalent formula without quantifiers. After elimination, each linear inequality is eventually constant and each congruence is periodic. Taking a common period h divisible by h_0 and a common threshold proves (4.18). The proximity bound and the bound obtained from the dual solution are effective, so the finite set of possible errors can be bounded before applying quantifier elimination. □

Comparing this slope with the fixing-number slope in Theorem 4.5 determines whether the cost of preserving color counts accumulates under repetition.

Corollary 4.12. *For a fixed graph G with $k = \chi(G)$,*

$$B(mG) - B_{\mathrm{w}}(mG) = m\big(\Lambda(G) - F_k(G)\big) + O_G(1).$$

The difference is bounded if and only if $\Lambda(G) = F_k(G)$. In that case it is eventually periodic; otherwise it tends to infinity linearly. The boundedness condition is decidable from the finite linear programs defining $\Lambda(G)$ and the finite definition of $F_k(G)$.

Proof. Subtract Theorem 4.5 from (4.17). When the two slopes agree, Theorems 4.6 and 4.11 give a common period for the difference. □

This gives a finite optimization criterion for bounded difference. A structural characterization of the graphs satisfying $\Lambda(G) = F_{\chi(G)}(G)$ remains a separate question.

5. Fixed-parameter recognition

The composition laws in Section 4 retain exactly the local information needed for repair, but those data can be large. We now address computation on arbitrary graphs. Small weak villainy forces a small vertex cover in the graph or its complement, so the relevant colorings can be described by boundedly many vertex types.

The classifications of graphs with $B(G) = 0$ in [2] and with $B(G) = 2$ in [12] give polynomial-time recognition algorithms for those two classes. We prove the stronger conclusion that recognition is fixed-parameter tractable in the threshold, for both repair rules.

Theorem 5.1 (Exact computation and recognition)**.** *There are computable functions f, h and an absolute constant c such that both $B(G)$ and $B_{\mathrm{w}}(G)$ can be computed exactly on an n-vertex graph in time*

$$h(B_{\mathrm{w}}(G))(n + 1)^c,$$

without either value being supplied. Moreover, given an integer $t \geq 0$, both

$$\min\{B(G), t + 1\} \qquad \textit{and} \qquad \min\{B_{\mathrm{w}}(G), t + 1\}$$

can be computed in time $f(t)(n + 1)^c$. Consequently, equality and threshold recognition for either parameter are fixed-parameter tractable in t, and belong to P *for every fixed t.*

The exponent c is independent of the threshold. The proof first forces a small vertex cover in the graph or its complement, and then uses different descriptions of repair in the two cases. The small-cover case uses effective quantifier elimination; no practical bound on f is asserted. Throughout this section, $\nu(H)$ denotes the matching number of a graph H.

5.1. **A structural consequence of a matching bound.** For an optimal coloring profile $\mathbf{m} = (m_1, \dots, m_k)$, where $k = \chi(G)$, put

$$p(\mathbf{m}) = \sum_{i=1}^{k} \left\lfloor \frac{m_i}{2} \right\rfloor.$$

This is the number of disjoint pairs of equal-colored tokens available in a word of profile $\mathbf{m}$. The next lemma extends the matching argument of Jahanbekam and Lin [12, Lemma 8].

Lemma 5.2. *For every optimal coloring profile* $\mathbf{m}$ *of* G*,*

$$B(G) \geq B_{\mathrm{w}}(G) \geq \min\{p(\mathbf{m}), \nu(G)\}.$$

Proof. Let $q = \min\{p(\mathbf{m}), \nu(G)\}$. Choose q edges of a matching and q disjoint pairs of equal-colored tokens. Place each pair at the ends of one chosen edge and distribute the remaining tokens arbitrarily. The resulting word has profile $\mathbf{m}$. Every proper coloring differs from it at an endpoint of each of the q selected edges. The edges are disjoint, so these discrepancies occur at distinct vertices. This argument does not require the repaired coloring to have profile $\mathbf{m}$, and therefore proves the bound for $B_{\mathrm{w}}(G)$. □

The matching bound forces a graph of small weak villainy to have either a small vertex cover or very many colors. In the latter case, the complement has a small vertex cover instead.

Corollary 5.3. *If* $B_{\mathrm{w}}(G) \leq t$*, then*

$$\nu(G) \leq t \quad \text{or} \quad \chi(G) \geq n - 2t.$$

In the first case G *has a vertex cover of size at most* $2t$*. In the second case* $\nu(\overline{G}) \leq 2t$*, and* $\overline{G}$ *has a vertex cover of size at most* $4t$*. The same conclusions hold if* $B(G) \leq t$*.*

Proof. Fix an optimal profile $\mathbf{m}$. Since every m_i is positive,

$$n - \chi(G) = \sum_i (m_i - 1) \leq 2 \sum_i \left\lfloor \frac{m_i}{2} \right\rfloor = 2p(\mathbf{m}).$$

If $\nu(G) > t$, Theorem 5.2 forces $p(\mathbf{m}) \leq t$, and hence $\chi(G) \geq n - 2t$.

The endpoints of a maximum matching form a vertex cover: otherwise an edge with both endpoints uncovered could be added to the matching. This gives the first cover bound. Moreover, a matching of size q in $\overline{G}$ supplies q disjoint nonadjacent pairs in G. Giving each pair one color and every remaining vertex its own color gives $\chi(G) \leq n - q$. Thus $\nu(\overline{G}) \leq n - \chi(G) \leq 2t$ in the second case, and the same endpoint argument gives its cover bound. Finally, $B_{\mathrm{w}}(G) \leq B(G)$. □

5.2. **Descriptions of colorings by types.** Let $U \subseteq V(G)$, with $|U| = s$, and suppose that $V(G) \setminus U$ is either an independent set or a clique. For a vertex $v \notin U$, define its type to be $\tau(v) = N_G(v) \cap U$. Write V_τ for the vertices of type τ and $n_\tau = |V_\tau|$, keeping only nonempty types. There are at most 2^s types. Any permutation fixing U pointwise and preserving each V_τ is an automorphism of G; let Γ be the group of these permutations.

For a finite label set L, describe a map $x : V(G) \to L$ by its restriction to U, called its *core map*, and by the vectors

$$\mathbf{x}_\tau = (x_{\tau i})_{i \in L}, \qquad x_{\tau i} = |V_\tau \cap x^{-1}(i)|.$$

Two maps have the same description if and only if an element of Γ takes one to the other. The next observation is the reason that descriptions can replace individual words in a repair problem.

Lemma 5.4. *Let* $\mathcal{A}$ *and* $\mathcal{C}$ *be descriptions of maps* $x, y : V(G) \to L$*. Define*

$$\delta(\mathcal{A}, \mathcal{C}) = |\{u \in U : x(u) \neq y(u)\}| + \sum_\tau \left(n_\tau - \sum_{i \in L} \min\{x_{\tau i}, y_{\tau i}\} \right).$$

For every fixed realization x *of* $\mathcal{A}$*, this is the minimum Hamming distance from* x *to a realization of* $\mathcal{C}$*.*

Proof. The labels on U are fixed. In V_τ, at most $\min\{x_{\tau i}, y_{\tau i}\}$ vertices can receive label i in both maps. These upper bounds can be achieved simultaneously: for each i, match that many occurrences of i in y to occurrences of i in the fixed map x, and then biject the remaining positions arbitrarily. Such bijections can be chosen independently in each type. The displayed formula follows by subtracting the maximum number of agreements from n. □

In particular, if a collection of permitted repairs is invariant under Γ, the minimum repair distance depends only on the description of the disturbed word. The minimum is computed by minimizing δ over the descriptions of permitted repairs.

5.3. **Graphs with a small vertex cover.**

Proposition 5.5. *Given a vertex cover U of size s, the values $B(G)$ and $B_{\mathrm{w}}(G)$ can be computed in time $g(s)(n+1)^{O(1)}$, for a computable function g.*

Proof. Put $I = V(G) \setminus U$. Since I is independent, $k = \chi(G) \le s+1$. We first compute k. For each $j = 1, \dots, s+1$, enumerate the proper maps $\varphi : U \to [j]$. Such a map extends to a proper j-coloring of G if and only if each nonempty type τ has a color outside $\varphi(\tau)$. Indeed, all vertices of a type can be assigned any one such color. The first successful value of j is k.

Direct enumeration of the type counts would give an exponent depending on s. Instead, we express the threshold condition by a short formula in Presburger arithmetic: integer linear inequalities and congruences with fixed coefficients, combined by Boolean operations and quantifiers. Effective quantifier elimination converts such a formula into an equivalent quantifier-free one. We use this established result, not its quantitative refinements; see Haase et al. [11, Corollary 3.2]. Universal quantifiers can be eliminated by negation.

Include all 2^s types, including empty ones. Introduce free variables $\mathbf{N} = (N_\tau)_{\tau \subseteq U}$ for their sizes and a free threshold variable ℓ. These variables are not yet assigned their input values. For an integer array $\mathbf{x} = (x_{\tau i})$ and a map $a : U \to [k]$, use the following predicates and linear forms:

$$\begin{aligned} R(\mathbf{x}, \mathbf{N}) :&\quad x_{\tau i} \ge 0 \text{ for all } \tau, i, \qquad \sum_i x_{\tau i} = N_\tau \text{ for all } \tau, \\ P_a(\mathbf{x}) :&\quad a \text{ is proper on } G[U], \qquad x_{\tau i} = 0 \text{ whenever } i \in a(\tau), \\ \mu_i(a, \mathbf{x}) &= |a^{-1}(i)| + \sum_\tau x_{\tau i}. \end{aligned}$$

Here properness of the fixed map a is a Boolean constant. The first predicate says that $\mathbf{x}$ describes a word; adding the second says that the word is proper. Write $\boldsymbol{\mu}(a, \mathbf{x}) = (\mu_i(a, \mathbf{x}))_{i=1}^k$.

Let $c, a : U \to [k]$ represent, respectively, an original coloring and a disturbance. The predicate

$$\begin{aligned} A_{c,a}(\mathbf{x}, \mathbf{z}, \mathbf{N}) :&\quad R(\mathbf{x}, \mathbf{N}) \ \wedge \ P_c(\mathbf{x}) \ \wedge \ R(\mathbf{z}, \mathbf{N}) \\ &\wedge \ \boldsymbol{\mu}(c, \mathbf{x}) = \boldsymbol{\mu}(a, \mathbf{z}) \end{aligned}$$

says that the disturbed description $(a, \mathbf{z})$ has a feasible profile, witnessed by the proper description $(c, \mathbf{x})$. For a target map $b : U \to [k]$, put $d_U(a, b) = |\{u \in U : a(u) \ne b(u)\}|$. Define $T_{a,b}(\mathbf{z}, \mathbf{y}, \mathbf{h}; \mathbf{N}, \ell)$ by the conjunction of

$$\begin{gathered} R(\mathbf{y}, \mathbf{N}), \qquad P_b(\mathbf{y}), \qquad \boldsymbol{\mu}(b, \mathbf{y}) = \boldsymbol{\mu}(a, \mathbf{z}), \\ 0 \le h_{\tau i} \le z_{\tau i}, \qquad h_{\tau i} \le y_{\tau i} \quad \text{for every } \tau, i, \\ d_U(a, b) + \sum_\tau N_\tau - \sum_{\tau, i} h_{\tau i} \le \ell. \end{gathered}$$

By Theorem 5.4, this predicate has witnesses $\mathbf{y}, \mathbf{h}$ exactly when a same-profile proper target with core map b is within distance ℓ. Indeed, the greatest possible total agreement in each type is $\sum_i \min\{z_{\tau i}, y_{\tau i}\}$, and is attainable for every fixed realization of the source. The variables $h_{\tau i}$ encode these agreements using only linear constraints.

Let $\Psi_a(\mathbf{z}; \mathbf{N}, \ell)$ abbreviate $\bigvee_{b : U \to [k]} \exists \mathbf{y} \, \exists \mathbf{h} \, T_{a,b}(\mathbf{z}, \mathbf{y}, \mathbf{h}; \mathbf{N}, \ell)$. Consider the formula

$$\Phi(\mathbf{N}, \ell) = \bigwedge_{c, a : U \to [k]} \forall \mathbf{x} \, \forall \mathbf{z} \ \left(A_{c,a}(\mathbf{x}, \mathbf{z}, \mathbf{N}) \implies \Psi_a(\mathbf{z}; \mathbf{N}, \ell) \right). \tag{5.1}$$

All quantified arrays range over the integers. After substituting $N_\tau = n_\tau$, this formula holds precisely when $B(G) \le \ell$. Every admissible disturbance has at least one proper profile witness, so the implication requires a repair for it. A disturbance with an infeasible profile has no such witness and imposes no condition. Although the witness is universally quantified, the permitted repairs depend only on the disturbance, not on which witness was used. For weak villainy, delete the profile equality from $T_{a,b}$ only; the source-feasibility predicate $A_{c,a}$ is unchanged. The resulting formula is equivalent to $B_{\mathrm{w}}(G) \le \ell$.

Both formulas have length bounded by a function of s: there are $k \le s+1$ colors, 2^s types, and at most k^{2s} pairs of core maps. All coefficients and core constants are bounded in s. Crucially, $\mathbf{N}$ and ℓ remain free during quantifier elimination. Thus the elimination time, the output length, and its coefficient lengths are bounded by a computable function of s, independently of the numerical type sizes. Substituting their actual values and evaluating the resulting formula takes $g(s)(\log(n+1))^{O(1)}$ time. This is a uniform algorithm: the formulas are generated from the given graph on U and then

processed by an effective elimination procedure. Binary search on $0 \le \ell \le n$, together with input processing, computes both values in the claimed time. □

5.4. **Graphs with a small complement cover.** Suppose next that $W = V(G) \setminus U$ is a clique. The chromatic number need not be bounded, but its computation still depends on only a bounded part of the graph.

Lemma 5.6. *If $|U| = s$ and $V(G) \setminus U$ is a clique, then $\chi(G)$ can be computed in time $2^{O(s \log(s+2))}(n+1)^{O(1)}$.*

Proof. Enumerate the partitions π of U into independent sets. In any proper coloring inducing π, a vertex of W either has a color not used on U, or shares a color with one part of π. No two vertices of W can share a color. Thus the possible sharing is described by a matching in the bipartite graph whose sides are π and W, with a part adjacent to $v \in W$ when v has no neighbor in that part. If a maximum matching has size $a(\pi)$, the fewest colors for this partition are

$$|\pi| + |W| - a(\pi).$$

Taking the minimum over π gives $\chi(G)$. There are at most s^s partitions for $s \ge 1$, and each matching is computed in polynomial time. The empty partition handles $s = 0$. □

We shall use the following elementary fact to handle the many singleton color classes. It does not depend on the structure of G.

Lemma 5.7. *Let $\mathbf{m}, \mathbf{m}'$ be positive profiles on $[k]$ of the same total size, and suppose that G admits a proper coloring of profile $\mathbf{m}'$. Let $J \subseteq [k]$ contain every color having multiplicity at least two in either profile. Replace each color outside J by a common label $*$. For a fixed word w of profile $\mathbf{m}$, the minimum Hamming distance to a proper coloring of profile $\mathbf{m}'$ equals the minimum Hamming distance between the resulting label maps.*

Proof. Different labels imply different colors, so passing to labels can only decrease a distance. To prove the reverse inequality, take any proper coloring y of profile $\mathbf{m}'$ and let R_w, R_y be the positions with label $*$ in the two maps. Both restrictions to the star positions are bijections onto the same color set $S = [k] \setminus J$, since every color in S has multiplicity one in both profiles.

Keep the colors of y outside R_y. On $R_w \cap R_y$, assign the colors of w. Fill $R_y \setminus R_w$ bijectively with the unused colors in S. This preserves the target profile and properness: all changed color classes are singletons. It also makes the two colorings agree at every position at which their label maps agree. Consequently its Hamming distance from w is exactly the label distance of the original pair. Minimizing proves the assertion. □

We can now combine the type description with Theorem 5.7: only the colors occurring more than once need individual names. The resulting count is controlled by the size of the cover and by the difference between the order and chromatic number.

Proposition 5.8. *Suppose that $|U| \le s$, $V(G) \setminus U$ is a clique, and $n - \chi(G) \le e$. Then both $B(G)$ and $B_{\mathrm{w}}(G)$ can be computed in time $f(s,e)(n+1)^{O(1)}$, for a computable function f. One may take*

$$f(s,e) = 2^{O(se+(s+e)\log(s+e+2))}.$$

Proof. Compute $k = \chi(G)$ by Theorem 5.6, and put $d = n - k \le e$. For every positive profile $\mathbf{m}$ on $[k]$,

$$\sum_i (m_i - 1) = d.$$

Consequently its set N of nonsingleton colors has size at most d, and

$$\sum_{i \in N} m_i = d + |N| \le 2e.$$

Up to a permutation of color names, there are only a bounded number of such profiles: their nonsingleton excesses $m_i - 1$ form a partition of d. A *profile shape* is a profile considered up to a permutation of color names. We enumerate these shapes, choose a labeled representative of each, and discard it if it is not the profile of a proper coloring. The properness test is included in the description enumeration below. A simultaneous permutation of color names preserves repair distances, so these representatives suffice for the outer maximum in both parameters.

First consider $B(G)$ and fix one such profile $\mathbf{m}$. Use labels $N\cup\{*\}$, with $*$ replacing singleton colors. The nonsingleton labels occupy at most $2e$ vertices. For a label map to come from a proper coloring, the vertices assigned each nonstar label must form an independent set. This can be checked from the description: its vertices in U must be independent, the label may occur at most once in the clique W, and any such vertex must have a type disjoint from that label's vertex set in U. The star positions impose no constraint, because they receive distinct singleton colors. We retain only descriptions having total count m_i for every $i \in N$ and total star count $k - |N|$, for both sources and targets.

By Theorem 5.7, the distance to a best repair of the same profile is exactly the distance to a best proper label map. By Theorem 5.4, this minimum, and then the maximum over disturbed words, can be evaluated using descriptions alone. The number of descriptions is bounded by a function of s, e. For example, after choosing the at most $(e+1)^s$ labels on U, the at most $2e$ nonstar occurrences outside U are distributed among at most $e2^s$ label–type pairs. An overcount is

$$(e+1)^s(e2^s+1)^{2e}.$$

Star counts are then determined by the type sizes. Testing all descriptions and comparing all pairs proves the first running-time bound.

For weak villainy, the target may have a different profile. Here the choice of target nonsingleton colors must be made after the disturbed word has been fixed. We describe this order explicitly. For each source description using labels $N\cup\{*\}$, choose one full-color representative w, assigning the source singleton colors bijectively to its star positions. All words represented by this reduced description have the same weak repair distance. Indeed, two such words are related by an automorphism in Γ followed by a permutation of singleton color names that fixes N. Both transformations preserve Hamming distances and the set of proper k-colorings.

With w fixed, let N' be the nonsingleton colors of a target. Colors in $N' \setminus N$ are singleton in the source; call them *promoted colors*. Each is identified by its unique position in w. Enumerating their names separately would introduce a factor depending on n with exponent e. We avoid this by using symmetries that fix the full source word.

Let $\gamma \in \Gamma$ permute only source-star positions within each outside type, fixing all other vertices. Define a color permutation π_γ by $\pi_\gamma(w(v)) = w(\gamma(v))$ on source singleton colors, and fix every color in N. For any target coloring y, put

$$(T_\gamma y)(v) = \pi_\gamma\big(y(\gamma^{-1}(v))\big).$$

Then $T_\gamma w = w$. Moreover, T_γ preserves properness and Hamming distance, because γ is a graph automorphism and π_γ is a permutation of colors. In particular, $d_{\mathrm{H}}(w, T_\gamma y) = d_{\mathrm{H}}(w, y)$.

These transformations act transitively on choices of promoted colors that occupy the same source singleton positions in U and have the same number of source positions in each outside type. Consequently, we need only choose a subset of the source singleton positions in U, and integers $h_\tau \geq 0$ with $\sum_\tau h_\tau \leq e$, not exceeding the available number of source-star positions of type τ. For each choice, fix one set of h_τ singleton names from that type. There are at most $2^s(2^s+1)^e$ such choices: the count vectors are overcounted by sequences of e type names or unused slots. Choose also the subset $N' \cap N$, and retain only sets with $|N'| \leq d$. For each resulting N', enumerate every assignment

$$m'_i \geq 2 \ (i \in N'), \qquad m'_i = 1 \ (i \notin N'), \qquad \sum_{i \in N'} (m'_i - 1) = d.$$

Every target is equivalent, under a transformation fixing w, to one of these choices. Its excesses are carried along by the color permutation and occur in the enumeration. For each choice, use the label set $J \cup \{*\}$ with $J = N \cup N'$. The star label now represents exactly the colors that are singleton in both profiles. Theorem 5.7 applies to this fixed source and target profile, and Theorem 5.4 computes the minimum over proper target descriptions. Colors in $N' \setminus N$ have one source occurrence, and colors in $N \setminus N'$ have one target occurrence; these occurrences retain their individual labels. Thus no potential agreement is lost by the reduction. Each target description must have total count m'_i for $i \in J$ and exactly $k - |J|$ star positions; target profiles with no proper description are discarded.

There are at most $2^{s+e}(2^s+1)^e$ choices of N', and a number depending only on e of assignments of its positive excesses. Each map on the union label set has at most $3e$ nonstar occurrences and at most $2e$ nonstar labels. Its descriptions are therefore bounded in number by

$$(2e+1)^s(2e2^s+1)^{3e}.$$

Minimize over the target profiles and descriptions for this fixed w, then maximize over source descriptions and admissible source profile shapes. The order of these operations is exactly the order in

Theorem 2.2. This proves the second running-time bound. The displayed description and orbit bounds, the number of profile shapes, and Theorem 5.6 give the stated choice of $f(s,e)$, with the cases $e=0$ interpreted directly. □

5.5. **The recognition algorithm.**

Proof of Theorem 5.1. Fix t. If G is empty, return zero for both parameters. Otherwise, compute a maximum matching in G. If its size is at most t, its endpoints form a vertex cover of size at most $2t$. Theorem 5.5 computes both parameters exactly; truncate each answer at $t+1$.

Otherwise $\nu(G) > t$. Compute a maximum matching in $\overline{G}$. If $\nu(\overline{G}) > 2t$, then $\chi(G) \le n - \nu(\overline{G}) < n - 2t$. Theorem 5.3 implies $B_{\mathrm{w}}(G) > t$, and hence also $B(G) > t$. Return $t+1$ for both parameters.

In the remaining case, the endpoints of a maximum matching in $\overline{G}$ give a set U of size at most $4t$ whose complement is a clique in G. Compute $\chi(G)$ using Theorem 5.6. If $\chi(G) < n - 2t$, the same dichotomy again gives both answers as $t+1$. If $\chi(G) \ge n - 2t$, apply Theorem 5.8 with $s \le 4t$ and $e \le 2t$, and truncate the exact answers at $t+1$.

This exhausts all graphs and proves correctness. By Theorem 5.5, the first branch takes time $g(2t)(n+1)^{O(1)}$, after replacing g by a nondecreasing computable bound if necessary. By Theorem 5.8, the complement-cover branch takes time $2^{O((t+1)^2)}(n+1)^{O(1)}$. Input processing and matching computations add only fixed polynomial factors. Their exponents are independent of t, proving the stated bound with a computable f. The argument includes $t = 0$.

To obtain the exact values with weak villainy as parameter, run this structural algorithm at thresholds $0, 1, 2, 4, \ldots$. Whenever an exact-computation branch is reached, retain both answers before truncation and stop. Continue only after a rejection, which certifies $B_{\mathrm{w}}(G) > t$. By Theorem 5.3, rejection is impossible once $t \ge B_{\mathrm{w}}(G)$. If $B_{\mathrm{w}}(G) \ge 1$, the largest tested threshold is at most $2B_{\mathrm{w}}(G)$; if it is zero, the first run succeeds. Taking a nondecreasing envelope of the parameter bounds and absorbing the number of runs proves the claim. □

Corollary 5.9. *Unless* P = NP*, there is no fixed integer t for which recognition of $B(G) = t$ is NP-complete. The same conclusion holds for $B_{\mathrm{w}}(G) = t$.*

6. Concluding remarks and questions

The composition laws quantify how global color counts couple components. For repeated copies, Theorems 4.6 and 4.11 give effective eventual formulas for both repair rules. Their slopes also decide whether the difference between the two values stays bounded. The next questions ask for structural descriptions and sharper bounds than the finite local optimizations provide.

Problem 6.1. Find structural or efficiently checkable versions of the criteria in Theorems 4.4 and 4.10. In particular, characterize the graphs satisfying $B_{\mathrm{w}}(G) = F_{\chi(G)}(G)$, equivalently those for which weak villainy is additive over every repetition.

For either parameter F, the *superadditivity deficit* is $F(G) + F(H) - F(G \sqcup H)$; a positive value measures a failure of superadditivity.

Problem 6.2. Determine sharp bounds on the superadditivity deficits at prescribed orders and chromatic numbers. For weak villainy, the unresolved deficit case has unequal chromatic numbers. Give a structural characterization of the graphs satisfying $\Lambda(G) = F_{\chi(G)}(G)$, the finite optimization criterion for bounded difference in Theorem 4.12.

The explicit stabilization bound in Theorem 4.6 comes from a general zero-sum estimate and need not be sharp.

Problem 6.3. Determine sharp bounds on the least stabilization index for weak repetition, meaning the least index from which the eventual periodic formula holds, and give graph-theoretic descriptions of its least period. Obtain comparable explicit bounds for the ordinary recurrence in Theorem 4.11.

The odd-cycle theorem of [13, Theorem 3.15] already gives singleton response families on a three-color palette, as discussed after Theorem 4.9.

Problem 6.4. For a fixed palette size, determine further graph classes whose response families in Theorem 4.9 admit polynomial-size representations supporting polynomial-time composition and queries. How much can these families be compressed without changing ordinary villainy after any further disjoint union?

The parameter dependence in Theorem 5.1 comes from general-purpose quantifier elimination. Can both values be computed by a direct combinatorial algorithm in time $2^{B_{\mathrm{w}}(G)^{O(1)}}(|V(G)|+1)^{O(1)}$?

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The authors used AI tools from OpenAI and Anthropic to assist with developing and checking arguments, computational experiments, literature searches, and manuscript preparation. The authors are responsible for all proofs, computations, and conclusions.

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